

locks', i.e., not using mutexes for a section of code that must be protected from concurrent execution.

Another well-known problem is 'open call', i.e., making a call to a method that is not thread-safe, from code that is part of a critical section.

There are other uncommon bugs as well, in this category. To give an example, the 'two stage access' problem occurs when a sequence of operations needs to be protected as a whole, but each operation is protected separately.

Liveness-related bugs

Deadlocks and livelocks are perhaps the best-known concurrency problems in this category.

A deadlock happens when there is a cycle in the resources acquired by different threads, and they hold on to the resources for ever (as it often happens at the traffic signals in India).

Livelocks happen when two or more processes continuously change their state in response to changes in the other threads without doing any useful work. For instance, one thread may create a file and another deletes

that file—and they keep watching for these events and are busy undoing each other's actions!

When high-priority threads keep using the CPU without letting lower priority threads do their tasks, we have the problem of starvation. So some work will never get done, and that will affect the program.

Sometimes, liveness problems happen because a thread 'waits forever'! For instance, I have seen an application hang because a thread that acquired a critical section never returned and the program was waiting for that thread to complete.

Concurrency bugs often take many hours to debug, and so it is better to be prepared and safe, than sorry. So, if you write concurrent programs, keep the three kinds of bugs mentioned in this article in mind and avoid them. You'll be happier for having done that. 

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